Georgios Is. Detorakis (GID), PhD

Contact Information

Lecturer in Neuromorphic Systems
Department of Computer Science
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Research Interests

Neuromorphic Computing: Development of neuromorphic frameworks and algorithms with applications in neuroscience and biodata analysis.
Machine and Deep learning: Algorithms for time-series analysis, forecasting, classification, stochastic neural networks, and physical-informed neural networks for solving PDEs and ODEs.
Neuroscience: Mathematical/computational modeling of cortical and subcortical brain areas, cortical plasticity, and neural fields theory. Combining deep learning and neuroscience methods to study reward mechanisms in the frontal cortex. Brain-machine interfaces with applications in neurodegenerative disorders such as Parkinson's disease.

Professional Experience

Lecturer in Neuromorphic Systems
University of Manchester | Manchester, United Kingdom — December 2025–Now
- Neuromorphic algorithms and computational neuroscience.
- Machine learning & deep learning algorithms for neuromorphic devices.
- Neuromorphic sensors.

Machine Learning Engineer
Independent Contractor | Irvine (CA), USA — December 2020-August 2025
- Develop machine learning algorithms with applications on time series analysis, forecasting, and classification.
- Deploy time series forecasting machine learning algorithms.
- Sentiment analysis for economic news.
- Deep learning computer vision applications on object detection & tracking, and image processing (super-resolution, denoising).
- Reinforcement learning for controlling TEC devices.

Data Science Architect
adNomus | San Jose (CA), USA — August 2019–December 2020
- Developed NLP algorithms with applications in context analysis.
- Analyzed behavioral data.
- Developed machine and deep learning algorithms for time-series prediction.

Research Experience

Postdoc Researcher in Neuromorphic Computing and Machine Learning
UCI | Irvine (CA), USA — January 2016–July 2019
Neuromorphic Machine Intelligence Lab
- Research in stochastic deep neural networks.
- Developed a neuromorphic framework (https://github.com/nmi-lab/NSAT).
- Developed machine learning algorithms for neuromorphic devices.
- Integrated neuromorphic sensors (DVS camera) with neuromorphic frameworks.

Postdoc Researcher in Neuroscience and Control Theory
CentraleSupélec | Gif-sur-Yvette, France — December 2013–December 2015
Laboratoire des signaux et systèmes
- Developed a mathematical model (neural fields with time delays) and a closed-loop simulation for Parkinson's disease treatment.
- Conducted theoretical work on non-linear retarded dynamical systems in a closed-loop setup.
- Co-developed software for spike-sorting (https://github.com/gdetor/SPySort).
- Developed Matlab software for on-line data processing on a Plexon recording device.

Research Assistant in Computational Neuroscience
Inria | Nancy, France — October 2010–October 2013
CORTEX Team
- Developed algorithms for self-organizing maps.
- Developed a mathematical model for the development of the somatosensory cortex.
- Studied attention mechanisms in the somatosensory cortex.
- Reproduced an *in vivo* experiment *in silico* to study self-organization in the brain.

Research Intern in Robotics
FORTH | Heraklion, Greece — May 2008–October 2008
Computational Vision and Robotics Laboratory
- Configuration and calibration of a HOAP3 humanoid robot.
- Applied biped locomotion algorithms on a HOAP3 robotic platform.

Research Intern in Systems Neuroscience
University of Crete | Heraklion, Greece — October 2007–April 2008
Laboratory of Systems Physiology and Computational Neuroscience
- Conducted EMG and EEG experiments for studying human tremor.
- EMG and EEG recordings, data processing and analysis.
- Software development for signal processing.
- Computational simulations of motor units.

Teaching Experience

Python Instructor
CentraleSupélec | Gif-sur-Yvette, France — March 2014–April 2014
- Taught a course on the Python Programming Language (Numpy, Scipy, Matplotlib).

Supervising
University of Manchester | Manchester, UK
- Undergraduate & Postgraduate students' projects in Computer Science.

Co-Supervising
CentraleSupélec | Gif-sur-Yvette, France — Fall Semester 2015
- Undergraduate students' projects in Engineering on oculomotor control.
  Primary adviser: Prof. Antoine Chaillet.

Inria | Nancy, France — Spring Semester 2013
- Postgraduate (M.Sc.) internship student working on sensorimotor integration.
  Primary adviser: Dr. Nicolas P. Rougier.

Tutoring
University of Crete | Herakleion, Greece — Spring Semester 2006
- Undergraduate course TEM 202 Theory of Algorithms.

Professional Service

Editorial Service
- The ReScience Journal
- Frontiers in Neuroscience–Neuromorphic Engineering

Workshop Service
- Co-organizer (with Prof. Antoine Chaillet) of a workshop on: "Neural Population Dynamics", Supélec, Gig-sur-Yvette, February 2015.

Education

Ph.D. in Computer Science
Inria/University of Lorraine, Nancy, France — October 2010–October 2013
- Thesis Topic: *Cortical plasticity, dynamic neural fields, and self-organization*.
- Area of Study: Computational Neuroscience.
- Adviser: Dr. Nicolas P. Rougier

M.Sc. in Brain & Mind Sciences
University of Crete | Heraklion, Greece — January 2007–January 2009

B.Sc. in Applied Mathematics
University of Crete | Heraklion, Greece — September 2002–September 2006
- Mathematical methods and software development track.

Awards & Grants

Travel Grants
- Federation of European Neuroscience Societies (FENS), Regional Meeting 2015, Thessaloniki, Greece (€500).
- Organization for Computational Neurosciences, CNS 2013 Annual Meeting, Paris, France ($ 200).

Reviewer Awards
- Outstanding reviewer for the years 2025 & 2023 for the IOP Neuromorphic Computing & Engineering.
- IOP trusted reviewer status in 2020.

Other Meeting Attendance

General Participant
- Workshop on *Neuromorphic Cognition Engineering*, Telluride, Colorado (USA), 2017.
- Summer school on *Neural Dynamics Approach to Cognitive Robotics III - A Hands–on*, Bochum (Germany), 2012.

Software

Software with applications in Neuromorphic Systems:

- **NSAT** A C/Python simulator for the Neural an Synaptic Array Transceiver (NSAT) neuromorphic framework (https://github.com/nmi-lab/NSAT).
- **NSATcarl** A C++ interface of CARLsim (http://www.socsci.uci.edu/~jkrichma/CARLsim/) with a neuromorphic framework (https://github.com/gdetor/CarlNsat).

Software with applications in Neuroscience:

- **Crebral** A simple C/Python simulator for conductance-based neural networks (https://github.com/gdetor/Crebral).
- **SPySort** A Python package for spike sorting (https://github.com/gdetor/spysort).
- **SI-RF-Structure** A collection of Python scripts that implement algorithms and experimental protocols for simulating area 3b of primary somatosensory cortex (https://github.com/gdetor/SI-RF-Structure).
- **SITopMaps** A Python/C implementation of self-organizing maps with applications on self-organization of area 3b of primary somatosensory cortex (https://github.com/gdetor/SITopMaps).

Software with applications in Optimization:

- **GAIM** A C++ library for Genetic Algorithms and Island Models (https://gitlab.com/gdetor/genetic_alg).

Communication Skills

Languages:
- Greek-native language,
- English-full professional proficiency,
- French-intermediate working proficiency.

Software Skills

Programming C, Python, C++, Rust, Fortran, Java, GNU Make, Pascal.
ML/DL Pytorch, TorchScript, TensorFlow, Ray, XGBoost.
PyData Ray, Scikit-learn, Pandas, Statsmodels, Numba.
Version Control Git.
Libs OpenMP, Nvidia CUDA, MPI, OpenCV, FEniCS, LAPACK/BLAS, Sundials.
Web HTML, CSS, Bootstrap.
Math Matlab/Octave, Maple.
Devops Gitlab CI/CD configuration.
NLP Hugging Face, spaCy.
Simulators Brian, Neuron (Python).
Editing Vim, TEX (LATEX, BibTEX, Tikz), Microsoft Office.
Graphics Graphviz, GIMP, Inkscape, Scribus.
OS Linux and BSD, Microsoft Windows family, Apple OS X.

Recording Techniques

Noninvasive:

- **Electroencephalography (EEG)** Conducted EEG recordings using an 18-electrodes portable EEG device.

Invasive:

- **Extracellular recordings** Set up, calibration and software development on a Plexon Recording Device.
- **Electromyography (EMG)** Conducted muscle force and motor units (MUs) activity recordings.

Expertise

Mathematics:

- Linear and Nonlinear Dynamical Systems, Control Theory, Numerical Analysis, Numerical Solutions of PDEs, Linear Algebra, Probabilities Theory, Theory of Systems and Signals, Information Theory, Optimization, and Empirical Dynamic Modeling (EDM).

Computer Science:

- Deep Learning, Machine Learning, Natural Language Processing, Neural Networks, Neuromorphic Computing, Software Development, Evolutionary Computing, Theory of Algorithms, and Topological Data Analysis (TDA).

Neuroscience:

- Systems Neuroscience, Computational Neuroscience, Cognitive Science, Neuroanatomy, Neurophysiology.

Talks

[1] "**Inherent weight normalisation in stochastic neural networks**"
University of Manchester (Inaugural Talk), 2026

[2] "**Biologically plausible contrastive divergence: Towards an abstract complementary learning system**"
Hughes Research Laboratory (HRL), 2017.

[3] "**Closed-loop deep brain stimulation for Parkinson's disease: A computational study**"
University of California Irvine, 2016.

[4] "**Neural Fields 101**", CentraleSupélec, 2015.

[5] "**The perception of touch: A computational approach**"
Aix Marseille University, 2014.

[6] "**Coherent 6–10 Hz rhythms in muscle activities-Humanoid Robot & Biped Locomotion-EEG & Time Series Analysis**"
Bernstein Center for Computational Neuroscience, 2008.

Submitted work

[1] E. Koiliari, G. Alevizopoulos, I. Mouzas, A. Vakis, O. Lesch, H. Walter, **G. Is. Detorakis**, and E. L. Pasparakis, *Promising efficacy of long-acting injectable antipsychotic medication in the treatment of community and institutionalized patients with co-occurring genetic neuropsychiatric diseases and alcohol dependence disorders*,
under review in Therapeutic Advances in Psychopharmacology.

[2] E. I. Koiliari, G. Alevizopoulos, A. Vakis, **G. Is. Detorakis**, I. Mouzas, H. Walter, O. Lesch, and E. L. Pasparakis, *Variables mainly determine the medical clinical course of patients with a dual diagnosis of psychosis and substance use disorder, after depot antipsychotic therapy, with a minimum treatment time of 6 months*,
under review in Psychiatriki.

Preprints/Technical Reports

[1] **G. Is. Detorakis** *Practical Aspects on Solving Differential Equations Using Deep Learning: A Primer*, arXiv:2408.11266, 2024.

Journal Publications (peer-reviewed)

[1] E. I. Koiliari, E. I. Kartsonas, **G. Is. Detorakis**, and E. L. Pasparakis, *Forensic psychiatric expertise based on articles 227/228 of Greek Law 4620/2019: A Descriptive series of fifteen cases*, Accepted for publication in Psychiatriki (Journal of the Hellenic Psychiatric Association), 2026. (Article in Greek)

[2] S. Dutta, **G. Detorakis**, A. Khanna, B. Grisafe, E. Neftci, and S. Datta, *Neural sampling machine with stochastic synapse allows brain-like learning and inference*, Nature Communications 13, 2571, 2022.

[3] R. Parise and **G. Is. Detorakis**, *OpenPelt: Python Framework for Thermoelectric Temperature Control System Development*, The Journal of Open Source Software, 7(73), 4306, 2022.

[4] N. P. Rougier and **G. Is. Detorakis**, *Randomized Self-Organizing Map*, Neural Computation, 33(8), 2021.

[5] **G. Detorakis**, A. Chaillet, and N.P. Rougier, *Stability analysis of a neural field self-organizing map*, The Journal of Mathematical Neuroscience, 10 (20), 2020.

[6] **G. Detorakis**, and A. Burton, *GAIM: A C++ library for Genetic Algorithms and Island Models*, The Journal of Open Source Software, 4(44), 1839, 2019.

[7] B. U. Pedroni, S. Joshi, S. Deiss, S. Sheik, **G. Detorakis**, S. Paul, C. Augustine, E. Neftci, and G. Cauwenberghs, *Memory-efficient Synaptic Connectivity for Spike-Timing-Dependent Plasticity*, Frontiers in Neuroscience (Neuromorphic Section), 13, 2019.

[8] **G. Detorakis**, T. Bartley and E. Neftci, *Contrastive Hebbian Learning with Feedback Random Weights*, Neural Networks, 114, 2019.

[9] **G. Detorakis**, S. Sheik, C. Augustine, S. Paul, B.U. Pedroni, N. Dutt, J. Krichmar, G. Cauwenberghs, and E. Neftci, *Neural and Synaptic Array Transceiver: A Brain-Inspired Computing Framework for Embedded Learning*, Frontiers in Neuroscience (Neuromorphic section) 12, 2018.

[10] N.P. Rougier, K. Hinsen, [et al., including **Georgios Detorakis**], *Sustainable computational science: the ReScience initiative*, PeerJ Computer Science 3, 2017.

[11] E. Neftci, S. Paul, C. Augustine, **G. Detorakis**, *Event-Driven Random Back-Propagation: Enabling Neuromorphic Deep Learning Machines*, Frontiers in Neuroscience 11, 2017.

[12] A. Chaillet, **G. Is. Detorakis**, S. Palfi and S. Senova, *Robust stabilization of delayed neural fields with partial measurement and actuation*, Automatica 83, 2017.

[13] **G. Is. Detorakis**, A. Chaillet, S. Palfi, and S. Senova, *Closed-loop stimulation of a delayed neural fields model of parkinsonian STN-GPe network: a theoretical and computational study*, Frontiers in Neuroscience, 9:237, 2015.

[14] **G. Is. Detorakis** and N.P. Rougier, *Structure of Receptive Fields in a Computational Model of Area 3b of Primary Sensory Cortex*, Frontiers in Computational Neuroscience, 8(76), 2014.

[15] **G. Is. Detorakis** *Cortical plasticity, dynamic neural fields, and self-organization*, University of Lorraine (Thesis), 2013.

[16] **G. Is. Detorakis** and N.P. Rougier, *A Neural Field Model of the Somatosensory Cortex: Formation, Maintenance and Reorganization of Ordered Topographic Maps*, PLoS ONE 7(7): e40257, 2012.

Reproducible Science (peer-reviewed)

[17] **G. Detorakis**, *[Re] A generalized linear Integrate-and-Fire neural model produces diverse spiking behaviors*, The ReScience Journal, 3:1, 2017.

[18] **G. Is. Detorakis**, *[Re] Multiple dynamical modes of thalamic relay neurons: rhythmic bursting and intermittent phase-locking*, The ReScience Journal, 2:1, 2016.

Conference Publications (peer-reviewed)

[19] **G. Is. Detorakis** and A. Papoutsi, *Neural Representations in Sequential & Interleaved Training*, Accepted for publication in SAB 2026 “From Animals to Animats 18” Proceedings.

[20] **G. Detorakis**, S. Dutta, A. Khanna, B. Grisafe, S. Datta, and E. Neftci, *Inherent Weight Normalization in Stochastic Neural Networks*, Advances in Neural Information Processing Systems (NeurIPS), 32, 2019.

[21] H. Kashyap, **G. Detorakis**, N. Dutt, J. Krichmar,and E. Neftci, *A Recurrent Neural Network Based Model of Predictive Smooth Pursuit Eye Movement in Primates*, 2018 International Joint Conference on Neural Networks (IJCNN), 2018.

[22] **G. Detorakis** and A. Chaillet, *Incremental stability of spatiotemporal delayed dynamics and application to neural fields*, 2017 IEEE 56th Annual Conference on Decision and Control (CDC), 2017.

[23] E. Neftci, C. Augustine, S. Paul, **G. Detorakis**, *Event-Driven Random Backpropagation: Enabling Neuromorphic Deep Learning Machines*, 2017 IEEE International Symposium on Circuits and Systems (ISCAS), 2017.

[24] B. U. Pedroni, S. Sheik, S. Joshi, **G. Detorakis**, S. Paul, C. Augustine, E. Neftci, G. Cauwenberghs, *Forward Table-Based Presynaptic Event-Triggered Spike-Timing-Dependent Plasticity*, 2016 IEEE Biomedical Circuits and Systems Conference (BioCAS), 2016.

[25] C. Pouzat and **G. Is. Detorakis**, *SPySort: Neural spike sorting with Python*, Proc. of the 7th Eur. Conf. on Python in Science (Euroscipy), 2014.

[26] N.P. Rougier and **G. Is. Detorakis**, *Self-Organizing Dynamic Neural Fields*, Advances in Cognitive Neurodynamics III, 2012.

Book Chapters

[27] A. Chaillet, **G. Is. Detorakis**, S. Palfi, and S. Senova, *ISS-stabilization of delayed neural fields by small-gain arguments*, In: Valmorbida G., Seuret A., Boussaada I., Sipahi R. (eds) Delays and Interconnections: Methodology, Algorithms and Applications. Advances in Delays and Dynamics, 10, Springer, 2019.

International Conferences

[28] E. L. Pasparakis, I. Mouzas, **G. Is. Detorakis**, and E. I. Koiliari, *Analysis of the therapeutic effect of the long-acting injectable form of Aripiprazole in incarcerated adult males in Greece*, EPA 2025, Madrid (Spain), 2025.

[29] E. I. Koiliari, I. Mouzas, **G. Is. Detorakis**, and E. L. Pasparakis, *The use of depot antipsychotics in patients with dual diagnosis of psychosis and substance use disorder is associated with improved quality of life, better general clinical outcome and fewer hospitalizations*, EPA 2025, Madrid (Spain), 2025.

[30] H.J. Kashyap, **G. Detorakis**, N. Dutt, J.L. Krichmar, E. Neftci *A neural network model of predictive smooth pursuit eye movement in primates*, SfN, San Diego (CA, USA), 2018.

[31] **G. Detorakis**, T. Bartley, E. Neftci, *Random Contrastive Hebbian Learning as a Biologically Plausible Learning Scheme*, OCNS, Seattle (WA, USA), 2018.

[32] **G. Detorakis**, T. Bartley, R. Parise, S. Sheik, C. Augustine, S. Paul, B. U. Pedroni, N. Dutt, J. Krichmar, G. Cauwenberghs, and E. Neftci, *Three-factor embedded learning on neuromorphic systems*, COSYNE, Denver (CO, USA), 2018.

[33] **G. Detorakis**, T. Bartley, R. Parise, S. Sheik, C. Augustine, S. Paul, B. Pedroni, N. Dutt, J. Krichmar, G. Cauwenberghs and E. Neftci, *Embedded Learning on Neuromorphic Systems: Towards a Unified Computing Framework*, NICE, Portland (OR, USA), 2018.

[34] **G. Detorakis**, T. Bartley, R. Parise, C. Augustine, S. Paul, E. Neftci, *Embedded learning on neuromorphic systems: Towards a unified computing framework*, IEED ICCAD HALO Workshop, 2017.

[35] **G. Detorakis**, D. Barsever, E. Neftci *NeuroLachesis: A Neuromorphic Framework*, Scipy 2017, Austin (TX, USA).

[36] A. Chaillet, **G. Is. Detorakis**, S. Palfi and S. Senova, *Robust stabilization of delayed neural fields by proportional feedback using input-to-state stability and small gain theorem*, ICMNS 2016, Juan-les-Pins, France.

[37] **G. Is. Detorakis** and A. Chaillet, *Closed-loop disruption of oscillations in a targeted frequency band for a delayed neural field STN-GPe model*, FENS Regional Meeting 2015, Thessaloniki (Greece).

[38] **G. Is. Detorakis** and A. Chaillet, *Incremental stability of delayed neural fields: a unifying framework for endogenous and exogenous sources of pathological oscillations*, CNS 2015, Prague (Czech Republic).

[39] **G. Is. Detorakis** and A. Chaillet, *Closed-loop regulation of the activity of delayed neural fields with only partial measurement and stimulation*, ICMNS 2015, Antibes - Juan les Pins (France).

[40] **G. Is. Detorakis** and A. Chaillet and I. Haidar, *A global stability analysis for delayed neural fields*, BCCN 2014, Göttingen (Germany).

[41] **G. Is. Detorakis** and N. P. Rougier, *A computational view of the primary somatosensory cortex*, CNS 2013, Paris (France).

[42] **G. Is. Detorakis** and N. P. Rougier, *Neural Fields and Cortical Plasticity*, Front. Comput. Neurosci. BCCN 2011: Computational Neuroscience and Neurotechnology Bernstein Conference & Neurex Annual Meeting, 2011.

Minor Conferences

[43] O. C. Giannouli, E.A. Petsalaki, A. Evangelou, **G. Is. Detorakis**, E. Froudarakis, and A. Papoutsi, *Cognitive Flexibility: Adapting Efficiently to a Dynamic Environment*, To Appear in the 15th FORTH Scientific Retreat, 2026.

[44] **G. Detorakis**, C. Augustine, S. Paul, E. Neftci, *Embedded learning on neuromorphic systems: Towards a unified computing framework*, 24th Joint Symposium on Neural Computation, San Diego (CA, USA), 2017.

[45] C. Pouzat and **G. Is. Detorakis**, *On the relation between neuronal size and extracellular spike amplitude and its consequence on extracellular recordings interpretation*, MathStat-Neuro Workshop, Nice (France), 2015.

[46] C. Pouzat and **G. Is. Detorakis**, *SPySort*, GDR Multielectrode systems and signal processing for Neuroscience, Gif-sur-Yvette (France), 2014.

[47] **G. Is. Detorakis** and N. P. Rougier, *Skin Topographic Maps in SI*, Progress in Neural Field Theory, Reading (UK), 2012.

[48] **G. Is. Detorakis**, N. P. Rougier, *Skin Topographic Maps in SI*, Workshop on Cognitive and Dynamics in Neural Systems: Mathematical and Computational Modeling (CONAS), Lyon (France), 2012.

Popular Science

[49] A. Chaillet, D. Da Silva, **G. Detorakis**, C. Pouzat, S. Senova., “Optogenetics to unravel the mechanisms of Parkinsonian symptoms and to optimize deep brain stimulation”, *ERCIM News, Special issue on cyber-physical systems*, Number 97, April 2014.